

# Runbook: Payment Gateway Regional Failover

Meridian Commercial Bank - Internal Document

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## Purpose

This runbook describes how to fail over Payments Gateway (PGW) traffic from the primary region to the secondary region during a latency or availability incident, and the first-line triage steps to run before deciding to fail over.

## When To Use This Runbook

- PGW p99 latency exceeds 1200ms for more than 5 minutes.
- PGW error rate (5xx) exceeds 2% for more than 3 minutes.
- Any PGW-related page during a card authorization incident.

## First-Line Triage (check before failing over)

1. Check CNA (Card Network Adapter) connection pool metrics in the payments-platform Grafana dashboard, panel "CNA Pool Utilization." If utilization is pinned at 100%, this is very likely a repeat of the connection-pool-exhaustion failure mode (see INC-2025-08-14-001 and INC-2026-05-11-005). Fixing the CNA config directly may resolve the issue without a full failover.
2. Check the deployments feed for any CNA or PGW deployment in the last 30 minutes. A recent deploy is the most common trigger.
3. Check downstream dependency health (compliance screening, fraud scoring) in the dependency status page. A downstream slowdown may be the true root cause rather than PGW itself.

## Failover Procedure

1. Confirm with the Payments on-call lead that failover is warranted (not required for a single-service config fix identified in triage).
2. Run: `pgw-cli failover --to-region secondary --reason "<incident-id>"`
3. Monitor the PGW dashboard for traffic shift completion (typically 2-4 minutes).
4. Verify synthetic transaction success rate returns to baseline (>99.5%) in the secondary region.
5. Post an update in `#incident-payments` with current status.

## Fail-Back Procedure

1. Once the root cause in the primary region is fixed and verified in a staging deploy, run: `pgw-cli failback --to-region primary`
2. Monitor for 30 minutes before closing the incident.

## Escalation

- Payments Platform on-call (PagerDuty schedule: payments-platform-primary)
- If unresolved after 30 minutes, escalate to Payments Engineering Manager.

- For customer-facing communication decisions, loop in Customer Support Lead.

**Related Documents**

- ARCH-payments-platform-overview
- INC-2025-08-14-001, INC-2026-05-11-005 (connection pool exhaustion incidents this runbook was updated in response to)